

Deep Learning Assignment 3

Melody-Conditioned Lyrics Generation using Recurrent Neural Networks

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1. Introduction

In this project we implement a recurrent neural network for generating song lyrics conditioned on melody information extracted from MIDI files. To investigate whether melody can influence lyric generation, three architectures were implemented and compared: a lyrics-only baseline, a model using global melody features, and a model using temporally aligned melody features. The models generate lyrics autoregressively by predicting the next word from the previous context. Additional experiments examined decoding strategies, melody influence, validation performance, and common generation failure modes.

2. Dataset

The dataset consisted of 600 training songs and 5 official test songs, each containing lyrics and a corresponding MIDI file. The small dataset size posed challenges for generalization and likely contributed to some of the generation errors observed later.

3. Preprocessing

Lyrics were preprocessed by converting all text to lowercase, removing punctuation, and replacing line breaks with a special <NEWLINE> token to preserve song structure. The text was then tokenized using whitespace tokenization. Special tokens were added for padding, unknown words, sequence start, and sequence end. The original vocabulary contained 7,702 unique words and was reduced to 4,153 after filtering rare terms. We used pretrained 300-dimensional GloVe embeddings (*glove-wiki-gigaword-300*), with out-of-vocabulary words initialized randomly, resulting in an embedding matrix of size (4153, 300).

4. MIDI Feature Extraction

MIDI features were extracted using the **pretty_midi** library. Two melody representations were used. The first was a 7-dimensional global feature vector containing statistics such as mean pitch, pitch variation, note duration, velocity, note density, tempo, and drum ratio. The second consisted of time-aligned melody features, where each MIDI file was divided into temporal segments and represented by local pitch, note count,

duration, and velocity features, producing a sequence of vectors with shape (30, 4). All melody features were normalized using statistics computed from the training set.

5. Model Architectures

Three recurrent neural network architectures were implemented and compared. The baseline model used 300-dimensional word embeddings as input to an LSTM followed by a linear output layer. The global melody model extended the baseline by concatenating a 7-dimensional global MIDI feature vector to each timestep, producing a 307-dimensional input. The aligned melody model instead incorporated a time-varying 4-dimensional melody vector at each timestep, resulting in a 304-dimensional input and allowing local melodic information to influence generation dynamically. All models were trained using the Adam optimizer with dropout, weight decay, and cross-entropy loss while ignoring padding tokens.

6. Training Configuration and Results

The models were trained using a sequence length of 30, batch size of 16, hidden size of 128, and learning rate of 0.001. The dataset was split into 540 training songs and 60 validation songs. This produced 14653 training samples and 1793 validation samples.

The baseline model achieved a best validation loss of approximately 4.9080. The global melody-conditioned model achieved a best validation loss of approximately 4.9403. The aligned melody-conditioned model achieved a best validation loss of approximately 4.9120. Training and validation loss curves for all models are provided in **Figures 1–4** in the appendix.

7. Decoding Strategies

Three decoding strategies were evaluated: proportional sampling, temperature-scaled sampling, and top-k sampling. Proportional sampling produced the most diverse outputs but often suffered from grammatical and semantic inconsistencies. Temperature sampling provided a balance between diversity and control, while top-k

sampling generated the most coherent and stable lyrics. A comparison of the output generated is provided in **Table 3**. Based on qualitative evaluation, top-k sampling ($k = 20$) was selected for all subsequent experiments.

8. Melody Influence Probe

To determine whether the melody-conditioned models actively used melodic information during generation we conducted a melody influence probe. The experiment kept the initial word and generation parameters fixed while replacing the correct melody with corrupted or mismatched melody inputs. Outputs were compared using Jaccard similarity and Jaccard distance, which measure vocabulary overlap between generated texts. Results are summarized in **Figure 5** and **Table 4**. For the aligned MIDI model, the Jaccard similarity was 0.296 (distance = 0.704), while the global MIDI model achieved a similarity of 0.320 (distance = 0.680). These results indicate that altering the melody substantially changed the generated outputs, suggesting that both models incorporated melodic information rather than ignoring it. The choice of initial word also

influenced the early content of the lyrics, although this effect weakened over time due to semantic drift.

9. Failure Modes

Three common failure modes were observed: degenerate repetition, grammatical incoherence, and semantic drift. The models frequently repeated high-frequency lyric patterns, generated syntactically inconsistent phrases, and gradually drifted away from coherent themes as generation progressed. These issues are characteristic of autoregressive language generation and were likely exacerbated by the relatively small training dataset. Representative examples are provided in **Table 5**.

Overall, the results indicate that melody information influenced lyric generation, with the aligned MIDI model showing the strongest sensitivity to melodic input. Despite coherence limitations, the models demonstrated meaningful integration of melody features. Detailed generation results for all test melodies and start words are included in Appendix B.

Appendices

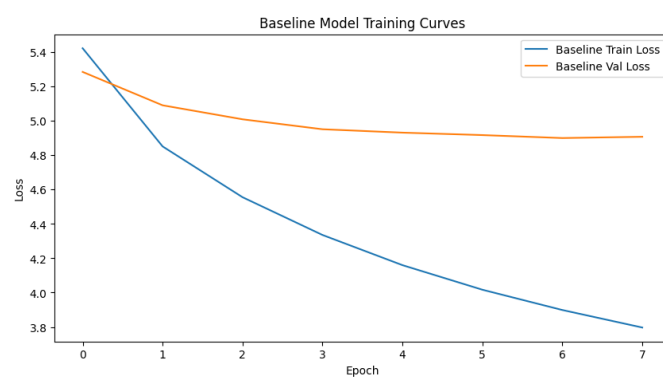


Figure 1: Baseline Model Training and Validation Loss

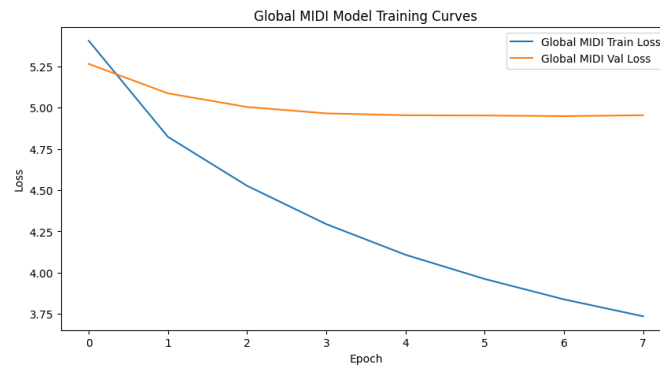


Figure 2: Global MIDI Model Training and Validation Loss

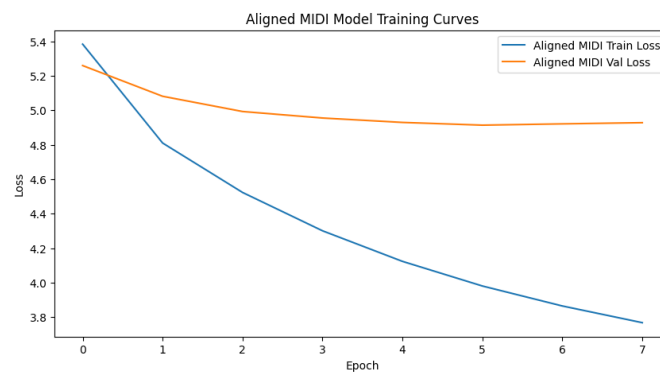


Figure 3: Aligned MIDI Model Training and Validation Loss

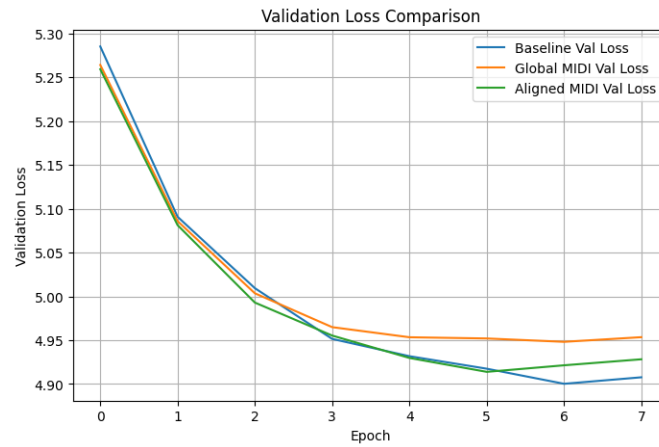


Figure 4: Combined Validation Loss Comparison

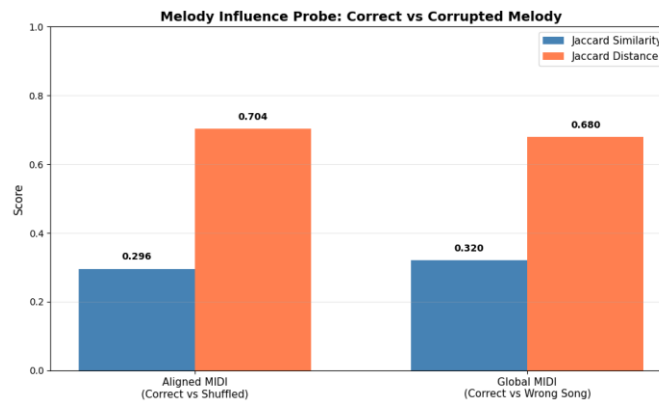


Figure 5: Bar chart comparing Jaccard similarity and Jaccard distance between lyrics generated with

correct versus corrupted melodies for both melody-conditioned models.

A Jaccard similarity of ~ 0.30 indicates that approximately 30% of vocabulary overlap exists between outputs generated with different melody inputs, while the distance of ~ 0.70 confirms substantial divergence. The aligned MIDI model (0.296 similarity) and global MIDI model (0.320 similarity) both demonstrate meaningful sensitivity to melody perturbations, with the aligned model showing slightly higher sensitivity

Table 1: Sample Outputs for Test Melodies (Eternal Flame, Start Word "love")

Model	Generated Lyrics
Baseline	love you say i know to say it is my name is a love i saw the heat you can tell me baby when she cried oh oh oh oh yeah

	i want to do i had i see i'll find the rain down i want to do it on fire that i'm gonna make it like that i ever been just what i am the heat
Global MIDI	love me i know that i feel for you can't you and i love i got and now oh and i never be so and i have to love to your love you and i'm like a song yeah you know i've done i can't you i was i love that i want to see but but i just a way oh oh
Aligned MIDI	love is the way my heart

	and the night fever if you i know how the good times you can the night i do it all i'm sure i want for a new attitude you if it is it you don't the things that you're the love cause it's hard your love in his we can do you you're no reason about a big
--	--

Caption: Generated lyrics for Eternal Flame by The Bangles with start word "love". The baseline model produces fragmented outputs. The global MIDI-conditioned model shows modest structural improvement. The aligned MIDI-conditioned model exhibits somewhat stronger local phrase structure, though all models remain substantially incoherent.

Table 2: Complete Table for All Test Songs (Single Start Word "love")

Test Song	Baseline (excerpt)	Global MIDI (excerpt)	Aligned MIDI (excerpt)
Eternal Flame	love you say i know to say it is my name is a love...	love me i know that i feel for you can't you and i...	love is the way my heart and the night fever...
Honesty	love my fantasy and i love you i will take the way...	love me alone just want to get home i want a little...	love divides will be with the right now oh you see...
Lovefool	love the day if i ever ever gonna get in a way...	love now it's not a new you i'm so...	love like an indication i just just a man...
Barbie Girl	love me and i'll give me so hard to get...	love you know you got you and let me i have...	love me on the way i'm gonna tell the heat...

All The Small Things	love you say what i'm the only one who...	love all my head i'm in your lips...	love and it all i could i just got...
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Excerpts from generated lyrics for all five test songs using start word "love" and Top-K sampling (k=20). The aligned melody model generally produces more locally structured outputs across songs, although grammatical and semantic inconsistencies remain present.

Table 3: Decoding Strategy Analysis, Comparison of Sampling Strategies

Strategy	Temperature	Top-K	Generated Lyrics (Excerpt)	Characteristics
Basic	N/A	N/A	love for new or getting join ask why do	High diversity, low coherence

		she to you have to your blues my love knew we is back feeling fever then all dial you are that feel at me i have faith you tear that know are the pop to write told her govern ment us your	
--	--	--	--

Tempera ture Sampling	0.8	N/ A	love you in the town at the one more i'm coming at the day we is a friend you i feel when you down into the best today i'm an old brick when there's it i listen to shoot and i underst and that fly i'd	Moderate diversity, moderate coherenc e
-----------------------------	-----	---------	--	---

Top-K Sampling	N/A	20	love if my life but i feel the blues but it's a life is it now that we can't be the sun i'm no one i can i feel the world i was a game the fire in a friend just don't need to you can do you see the	Lower diversity, highest coherenc e
----------------	-----	----	---	---

Comparison of three decoding strategies using the aligned melody-conditioned model. Top-k sampling

(k=20) produced the most coherent output and was therefore used for the main evaluation experiments.

Table 4: Quantitative Results of Melody Perturbation Experiment

Condition	Jaccard Similarity	Jaccard Distance
Correct Melody vs Shuffled Melody (Aligned MIDI)	0.296	0.704
Correct vs Wrong Melody (Global MIDI)	0.320	0.680

Correct Melody:
love is
you know what i am
so won't my baby
i say i've tried to see
and you to make all the world
that i don't
don't make
now you

it's too
if a game
that was a love me oh oh woah
oh oh

Shuffled Melody:
love will it
i will all my heart
i know it's the day
with my heart
oh yes i'm so all your baby
when
i do you can't be i do i
need that i've been a way i'm goin'
the world baby
so many time you
if we can show

Melody perturbation experiment comparing lyrics generated with the original melody versus a shuffled melody. The Jaccard similarity of 0.296 and distance of 0.704 indicate substantial divergence, suggesting that melodic information influences lyric generation.

Table 5: Representative Failure Mode Outputs

Failure Mode	Example from Output	Hypothesized Cause
Degenerate Repetition	love for new or getting / join ask why do she to you have / to your blues my love	Repeated reliance on high-frequency lyric tokens without semantic progression.
Grammatical Incoherence	i have faith / you tear / that know are the pop to write told / her government us	Failure to maintain grammatical structure across longer spans.
Semantic Drift	love you in the town / at the one more / i'm coming at the day we / is a friend	Accumulation of prediction errors causes drift away from coherent themes.

Appendix B: Complete Generated Lyrics for Test Melodies

=====

Artist: the bangles

Title: eternal flame

Start word: love

--- Baseline ---

love you say i know to say it

is my name is a love

i saw the heat

you can tell me baby when she cried

oh oh oh oh yeah

i want to do i had

i see

i'll find the rain down

i want to do it on fire

that i'm gonna make it like that i
ever been just what i am the heat

--- Global MIDI ---

love me
i know that
i feel for you can't you and i
love i got
and now
oh
and i never be so
and i have to love to your love
you and i'm like a song yeah you
know i've done
i can't you
i was

i love that i want to see

but

but i just a way

oh oh

--- Aligned MIDI ---

love is the way my heart

and the night fever

if you

i know how the good times

you can the night

i do it all i'm sure

i want for a new attitude

you

if it is it

you don't the things that you're the love

'cause it's hard your love in his

we can do you

you're no reason about a big

=====

Artist: the bangles

Title: eternal flame

Start word: night

--- Baseline ---

night is a big it was so long

ago i see

i know i saw my love can you

make me

you were the stars in our hearts and

you can't make it
i'm in the fire
and the end my mind of my fantasy
i don't want it on the world
you need to me
i need some to make me i have
you say my heart

--- Global MIDI ---

night
the day
oh yeah
just can't believe the time i'm gonna have
i need you can't
and you
if i can be my heart in your

love will see you see it
it's true it's got you can
i want it
and a face
and i'm sorry yeah oh no more
you
and me a woman in me
just so much is you

--- Aligned MIDI ---

night
if i am i won't you can give
me
i'm gonna get
i wanna make my best die
and i don't need your baby

you to you don't be a love that
they can find
when they were the love to find you
can't it a day the night
and i could know can tell me don't
let me the night
you don't
i can make a little

=====

Artist: the bangles
Title: eternal flame
Start word: baby

--- Baseline ---

baby
we go
you know you can make my dreams
you're no
if i see you can take it
so sweet
you're in my oh oh
and love don't make it is it is
all alone
the one
so bad
the world
so far that you're gonna get away to
find
and it all the way all you can't
be

the only

--- Global MIDI ---

baby

and we want for me

you

and the only more more a time to

your love somebody to the love me to

the morning

what you

to me

i will find to find you love to

make it was just just can't just my

love you

i could i can't see that i want

to me

and i can it would

i can ever

and it's

--- Aligned MIDI ---

baby

the way to you feel like no way

it all that i could be love is

a day

i'm sure you

and i see

and that's my love in time

when the sun

oh yes

if you feel so long

the stars or my life and i will

never will all over the same

to

we got out there the good time

=====

Artist: billy joel

Title: honesty

Start word: love

--- Baseline ---

love my fantasy

and i love you

i will take the way

and then i ever knew no way

you can't be at all the rain

you say
it's the way to dance
i want to do it
i'm not a wild
i'm so
i can't make this
it's gonna try me
so
i see that the night
the sky
i will

--- Global MIDI ---

love me alone just want to get home
i want a little
but you

'cause i see
i want of the love can come through
it the way
and
if
'cause i can't stop
you know
we can do it just and if you
know i'm a white
you
the light
i would ever you
i can do
so
and i can't you

--- Aligned MIDI ---

love divides will be with the right now

oh you see you see you say it's

too much time

i'm in a game all of this world

my heart to dance

so don't make

so many times

but you see

the things of my face the night

i never had me

and i won't see i'm in my soul

and he we have to dance baby

that we

=====

Artist: billy joel

Title: honesty

Start word: night

--- Baseline ---

night

and the way

and i will be the stars of all

you got and you will be afraid of

this girl

and

'cause if you can be

i'm only yesterday

i never ever wanted to see and

i can i can't
if at your hair and when you're not
it would
i can you don't make it was the
end your eyes
and the way
i

--- Global MIDI ---

night
the sun
i'm just can't see
i know to me
i'm gonna love divides of me and i
love is it seems
i would you see

i love me to me and i love
to say
i need this before my love you
i got up
i can the best go away
you
the world
we can make to you
i want to

--- Aligned MIDI ---

night
but i'm gonna be a dream
and i don't want to the sun will
'cause i am
i can give

oh oh
you can you can i can give me
and they say that i'm sorry
the love to get
the fire
when you can't take this
you're the night
the way on your side up to have
a woman
when the world and

=====
=====

Artist: billy joel

Title: honesty

Start word: baby

--- Baseline ---

baby you make your love you know i'm

not you can be a whole

a little

you do we go with your love to

say it

and i don't see me

and when i'm gone for your body baby

it's a game

if my heart

the fire

and if i don't you come down in

town

and i'll be

i know

oh oh

--- Global MIDI ---

baby

we need to love in hell

'cause we got to see you see the
world

to see it

you know how i know we have

i'm my mind i have to me alone

that i never stop to you don't like

i would it seems

i want to find me

you're a song

you're

and i'm the sign me all my soul

that i

--- Aligned MIDI ---

baby baby

i want the night

'cause i can be

you want to make it is it

i'm your head

if

the world

you say you

i want to the best

and the best

and it's my mind in the sun and

i will

i'm gonna live

that i could find in the world

you see

i'm falling and that's my love you

=====

Artist: cardigans

Title: lovefool

Start word: love

--- Baseline ---

love the day

if i ever ever gonna get in a

way

you

'cause i'm so

and i had to me
i can you got in this dream to
be
you want to me
if it
my own on the way i want some
it
i don't see the way
we come on the rain and i see
and i was
it's not it
i got

--- Global MIDI ---

love
now it's not a new you

i'm so
it's not i could be
and the good times
it's not a new
i'm gonna get the love like so far
the first
that i get into out of the the
end is what to find a chance to
say it a face a fool
that you just don't i don't you just
can you don't

--- Aligned MIDI ---

love like an indication
i just just a man
i am

you need to be you got
but
and you
and you see and me for a little
prayer
it's too many days time
i could i am i'm not a river
to you
when you
i feel all that would be
you're not you can a dream in your
head your hand you
the

=====

=====

Artist: cardigans

Title: lovefool

Start word: night

--- Baseline ---

night

but it's it it all the world

that i've no one day to see

oh

don't and i see your body

'cause

and it's gonna do my fantasy

but

it's no son

that i got in her name and

you say it's not a little respect you

can you can see

it's no one

oh

'cause i will make it

--- Global MIDI ---

night in

just the only the night

it's only ever to me

i was dry

that your mind

but you can you

that you're gonna try

you

if you just can bring you make it

i know you can love you know

if a song that i got no is
i want to be
i'm so i feel like an invisible

--- Aligned MIDI ---

night
the sun will be the end
you
i can do you
if i will you see in the way
now
when you're a new game my love is
the night
you
but i'm so good
if you

i think it
but you're all the time the end
and it's a love can see i'm looking
at least
i could be the good that

=====

Artist: cardigans

Title: lovefool

Start word: baby

--- Baseline ---

baby my soul don't you say that it
i'm gonna be
so

when i got to find
'cause if i don't give you can you
know
i just can't go
i'm your hand and the love
i have to me for you see you
come on to the fire
but i can do
i could make me
a way to see
i want it don't know

--- Global MIDI ---

baby baby i was the the only love
now
so hot lady

they have a way on his heart
'cause that the way
so
i get it
don't just a fool a man the one
who wants to me
i have to do it my baby doll
i was
'cause
i have you can be wrong i feel
just don't be

--- Aligned MIDI ---

baby
if you
'cause i'm so long

i'm sorry that i can you feel your
soul brother i have to the right
that was just all in the stars that
was so
the same
i could you see
we need me now
i was the stars in the sun
to a man
if your love you can you
if you were the day

=====

Artist: aqua

Title: barbie girl

Start word: love

--- Baseline ---

love me and i'll give me so hard

to get

i'm in in in the moon is so

i want it

you say

oh oh i'm not a chance to take

her mother

we go

it's it is it

so

don't get so many

that you come i feel

so

a man

and i'll be so don't want the one

it's so

--- Global MIDI ---

love you know you got you

and let me i have

don't you see you know

i never can tell you can't you need

your heart is the way my way for

the heat for a little

and the the morning the way the law

won it

i feel

i'm gone the world

don't do it

don't think

i think it's in the law won

--- Aligned MIDI ---

love me on the way

i'm gonna tell the heat

with me don't get to see

but i know you and it

you're the best in in the world

and the good timin' but if i see

that we got friends the end of me

i'm not

you're trying

oh

i'm the sun

and you

so you do

i'm so

oh

and i

=====

Artist: aqua

Title: barbie girl

Start word: night

--- Baseline ---

night

oh yeah yeah and the way

the love to know where we are here

that i feel like the same

you
i can i had your eyes
i could never have i need your side
i'm a face
if it
and now i don't you want
if i never had just you see i
can you don't get my heart to see
i got

--- Global MIDI ---

night
and the the things i just so so
long
i can't take it
a love can do to take it was

a way a chance the one
and say you're looking into this i know
that you feel to me
and i will the love
if i don't want to make me
so i'm so don't gonna get my
the sun and

--- Aligned MIDI ---

night
and let me
and i want your desire
they say if i don't get
it's a place my head in the sun
now i've been for a beautiful
i don't do

'cause we're nobody i'm gonna tell you got
on the time for you
it's not your head in the way it
i'm sorry my way on the music they
it's a beautiful oblivion

=====

Artist: aqua
Title: barbie girl
Start word: baby

--- Baseline ---

baby girl
i can't find a man is your body
baby oh and my own a man

and my
you don't be
you can't go
i'm in my name is so long and
i know
that the way
i know i don't get a whole
i'm gonna be the end to the sky
is all i know
i never knew
and now to see
i

--- Global MIDI ---

baby oh yeah
so all

i will love to take her
you're
don't be with you
and i can you think that a little
girl
i can't do
the rain
don't leave the rain i want a love
i can do
but that i was your soul
and you don't want me and i have
been a way all and the day is

--- Aligned MIDI ---
baby baby don't be a man in my
love

we have
so
but if your life
now you
with his
and let me
i can i can do me
it's hard to do to the night
don't know that you
now
if it's time
that i'd wait all the way you will
i'm here that
i won't come down
i say to find

=====

Artist: blink 182

Title: all the small things

Start word: love

--- Baseline ---

love you say what i'm the only one

who

i can say i don't want to hear

me

the moon in the end to stay

my head

you

i got to say to hear the moon

and i will it don't want to hold

you have a new of me the way
to know i got the world
we know and i am whatever it

--- Global MIDI ---

love all my head
i'm in your lips
and i just you can't the best on
a good times
i'm waiting out on on your
the law i gotta stop to the law
in the law i am
oh yeah yeah
i love you got all in your life
knows
if you don't need to me

and i have got it all right

--- Aligned MIDI ---

love and it all i could i just

got

i can't take me

i see the way the time

i feel like a friend the heart it's

a place the world

all my face

that is the same

i have to see a bottle up

let a little piece little

and you i am

you don't do what the law

=====

Artist: blink 182

Title: all the small things

Start word: night

--- Baseline ---

night

i can be

oh i have no more so hard to

do to me

oh

a man in the end

i'm gonna get and i can't be

the one and i'm not you don't just

like a game and

and i don't be the way i am

oh

if your

that you don't do you got to hold

her

my life on broadway

i ever

--- Global MIDI ---

night the country so much you're you can

know how i've done

i need

and put the morning and it's my mind

that you're just wanna give you gotta see

you got a chance on the funk soul

i was with the law it all the

country the things it
i need you see the law in the
law you're on the first a

--- Aligned MIDI ---

night
when i can a man to me feel
so so
and i'm holding a new place i want
you see the good the way
when i'm in her body
i was there i will be my baby
you
i am whatever
i wanna talk
and it's just a girl

and
so long
i'm on the end
if it's got so sweet
the good

=====

Artist: blink 182
Title: all the small things
Start word: baby

--- Baseline ---

baby
it's no
i don't get on the world the night

the sun in my oh oh yeah that
you're your mind i don't i feel in
time
the rain
and then it
i feel all that you can't all your
love you say it was so far more
i love don't let you see my own
oh
but she
i'll take your face me
but

--- Global MIDI ---

baby baby it's my mind that the the
law won

they have a little hot
that's that i see
now the funk soul yeah
you want
i know why you
and you
if you
i can't do to me
she's got to the night
i think i want to me a man
so much
if you're the funk soul in your mind
on

--- Aligned MIDI ---

baby

and that's you can't take it
i don't you need a dream
it's not
don't get you know you can't understand
the night
you don't like
'cause you
the world
the way
'cause i see you
when i just just the way the love
divides want to me
i want to my love
you're all i see i got that